Check for updates

Biotechnology and Bioengineering

WILEY

BIOTECHNOLOGY AND BIOENGINEERING

MINI REVIEW

Artificial Intelligence-Powered Algal Biodiesel: The Future of Biofuel Production Through Data-Driven Biotechnology

Gyanendra Tripathi

Department of Bioengineering, Integral University, Lucknow, Uttar Pradesh, India

Correspondence: Gyanendra Tripathi (gyanan2803@gmail.com)

Received: 4 March 2026 | **Revised:** 27 June 2026 | **Accepted:** 29 June 2026

Keywords: algal biodiesel | artificial intelligence | deep learning | hybrid models | machine learning

ABSTRACT

Microalgae are promising feedstocks for sustainable biodiesel production, although their industrial use is currently limited by their biological complexity and high production costs. Recently, artificial intelligence (AI) technologies have been identified as promising, game-changing tools that could optimize algal biodiesel processes by leveraging data-driven biology. The purpose of this Review is to provide an integrated summary of recent developments in machine learning, deep learning, and hybrid models of microalgal biodiesel, analyzing AI frameworks that integrate biological and physical information with microalgal biodiesel process models to improve productivity, biological stability, and sustainability. It is essential to critically analyze the challenges of data quality and explainability in machine learning models, and to explain their use in techno-economic analysis of microalgal biodiesel. In the final part, directions for developing an independent microalgal biorefinery system using artificial intelligence tools are outlined to accelerate the lab-to-production transition of climate-relevant biofuels.

1 | Introduction

The quest for sustainable liquid fuels worldwide has established microalgae as one of the most promising biological models for biodiesel production. The use of microalgal systems offers numerous benefits over land plants for oil production, including fast growth rates, high lipid productivity, no competition for food, and the ability to be grown on non-arable land using industrial CO_2 (Bedru et al. 2025). These benefits align closely with climate management efforts to prevent global warming, making microalgal biodiesel a highly desirable component of the future renewable fuel mix. Nevertheless, despite years of research, the commercialization of microalgal biodiesel has not been a major factor. The complexity of algal systems has prevented successful scaling-up efforts from following laboratory successes. This is primarily because the rates of lipid expansion and algal cell growth involve non-linear interactions that affect genetic traits and operational characteristics. Divergent variations in algal cell metabolisms and production rates may result from minor disruptions brought on by changes in temperature, nutrients, and light intensity (Xue et al. 2021; Tang et al. 2025; Mohapatra et al. 2025). The complexities of multidimensional operations cannot be effectively handled by conventional optimization tools that focus on experimental procedures and single-variable optimization. The supply chain for algal biodiesel also comprises several intricately linked process stages, including strain selection, growth, harvesting, lipid extraction, and fuel production. Process efficiency, energy ratio, overall sustainability, and downstream process steps are all influenced by modeling decisions made in these initial stages. Highly diverse data from biological, chemical, and engineering perspectives cannot be smoothly combined by conventional models (Alber et al. 2019; Tang et al. 2026; Yusof et al. 2025). Artificial intelligence (AI) has become a ground-breaking strategy for biotechnology innovation in this regard. AI has enabled the discovery of patterns and the forecasting of system responses that are not achievable with conventional techniques, and it has

© 2026 Wiley Periodicals LLC.

the potential to analyze and train on data from complex, high-dimensional systems. Drug development, synthetic biology, and bioproduct processing are just a few of the applications that have been transformed by recent developments in machine learning (ML) algorithms and model integration (Gupta et al. 2025). AI has the potential to disrupt the algal biodiesel industry due to the industry's high data and system complexity (Saju et al. 2025; Deepankumar and Kl 2026).

Recent advances in algal biotechnology have brought AI to the forefront as a powerful tool for process optimization; however, the available literature lacks a systematic review of specific AI applications. This overview provides an overview of various AI uses for microalgae-related processes and their efficiency. This review provides a systematic overview of AI approaches used at all stages of biodiesel production from algae. A major advantage of the current research is the development of a structured framework of modeling approaches based on three main types of models—data-driven, mechanistic, and hybrid. Each model is characterized by performance, scalability, data demand, and interpretability. Besides, this study presents not only a qualitative analysis of existing research but also a quantitative synthesis of the results. It provides additional information on techno-economic analysis (TEA) and life cycle assessment (LCA). The study also focuses on the potential and limitations of implementing autonomous algal biorefineries.

Several recent reviews have considered the use of AI in microalgae cultivation, metabolic engineering, and biofuel production. Nevertheless, most of them have focused on a single aspect of the production chain, for example, strain development, growth enhancement, or machine-learning applications for predictive modeling. This review is distinct from others in its holistic approach, incorporating not only data-driven, mechanistic, and hybrid modeling but also applying them across the entire value chain of algal biodiesel. In contrast to other reviews on the topic, the current article integrates the evaluation of models' performances with TEA, life-cycle assessment (LCA), issues in scaling up and commercializing algal biofuel systems, explainable AI (XAI), and futuristic ideas for fully autonomous biorefineries.

2 | AI in Biotechnology- Concepts Relevant to Biofuels

AI is a broad class of computational methods, usually categorized by their focus on knowledge extraction and mining, and on tasks such as prediction, classification, and decision-making. In biotechnology, there is an emerging trend toward the use of AI, as experimental methods yield complex, high-dimensional data that traditional statistical models struggle to analyze. AI is evolving to complement traditional models in understanding complex systems with nonlinear dynamics by establishing connections between variables using data rather than mathematical equations (Karaca and Baleanu 2022; Amina Raza 2024).

ML, the most popular field of AI, enables algorithms to identify patterns in labeled or unlabeled datasets. Supervised models such as regression, random forests (RFs), and support vector machines (SVMs) have been widely used to predict biological outcomes, including growth rates and biological performance (Freitas et al. 2025; Gerdan Koc et al. 2023). Unsupervised models, such as clustering and dimensionality reduction tools, are useful for identifying unknown biological states, strains, and metabolic responses (Freitas et al. 2025). These are largely beneficial for algal models, as they are so diverse and responsive to their environments that they cannot be distinguished using standard classification methods. Deep learning (DL) extends ML algorithms by using multi-layered neural networks that can easily capture nonlinear relationships (Alzahrani et al. 2025). In biotechnology, the application of DL has demonstrated excellence in image analysis tasks and in interpreting sequences through sensor monitoring. In algal biodiesel production, convolutional neural networks (CNNs) can be used to automatically analyze cell morphology, while recurrent and transformer models can be useful for modeling growth patterns. Such models can easily run near-real-time analyses (Kuila and Kumar 2025). Beyond predictive analytics, reinforcement learning offers another framework that elevates decision-making. This includes algorithms that pick up tactics through interactions with systems or the environment. Reinforcement learning can be used to schedule nutrient applications and create self-optimized growth conditions for biofuel production. These can be followed by digital twins, low-cost bioprocess simulations that can explore parameter spaces. In the field of biotechnology, hybrid modeling paradigms that combine AI techniques with mechanistic or first-principles models are becoming increasingly common. These techniques leverage data-driven learning while capitalizing on the simplicity and interpretability of biochemical kinetic and mass-balance principles. Because hybrid modeling addresses biological uncertainties and allows biological knowledge to inform AI predictions, it is particularly attractive for modeling algal biodiesel processes (Tang et al. 2026; Freitas et al. 2025; Demirel and Rosen 2023; Riezzo et al. 2025; Tsopanoglou and Jiménez del Val 2021).

Notably, the successful application of AI in biofuels depends on the knowledge domain and data quality in addition to high-quality algorithms. Biotechnological applications often use noisy, sparse, and/or domain-specific datasets, necessitating careful data verification and interdisciplinary collaboration. Designing intelligent and scalable algal biodiesel platforms and evaluating modern applications requires an understanding of the aforementioned basic AI concepts (Deprá et al. 2024; Santos et al. 2025) (Figure 1).

The AI model for algal system dynamics can be formulated mathematically as a set of nonlinear function approximations. For example, the conventional feed-forward artificial neural network (ANN) model can be described as follows:

$$y = f\left(\sum_{i=1}^{n} w_i x_i | b\right),$$

where x_i is the input vector (e.g., irradiance, nutrient concentration), w_i is the synaptic weight, b is the bias, and $f(x)$ is the non-linear activation function.

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

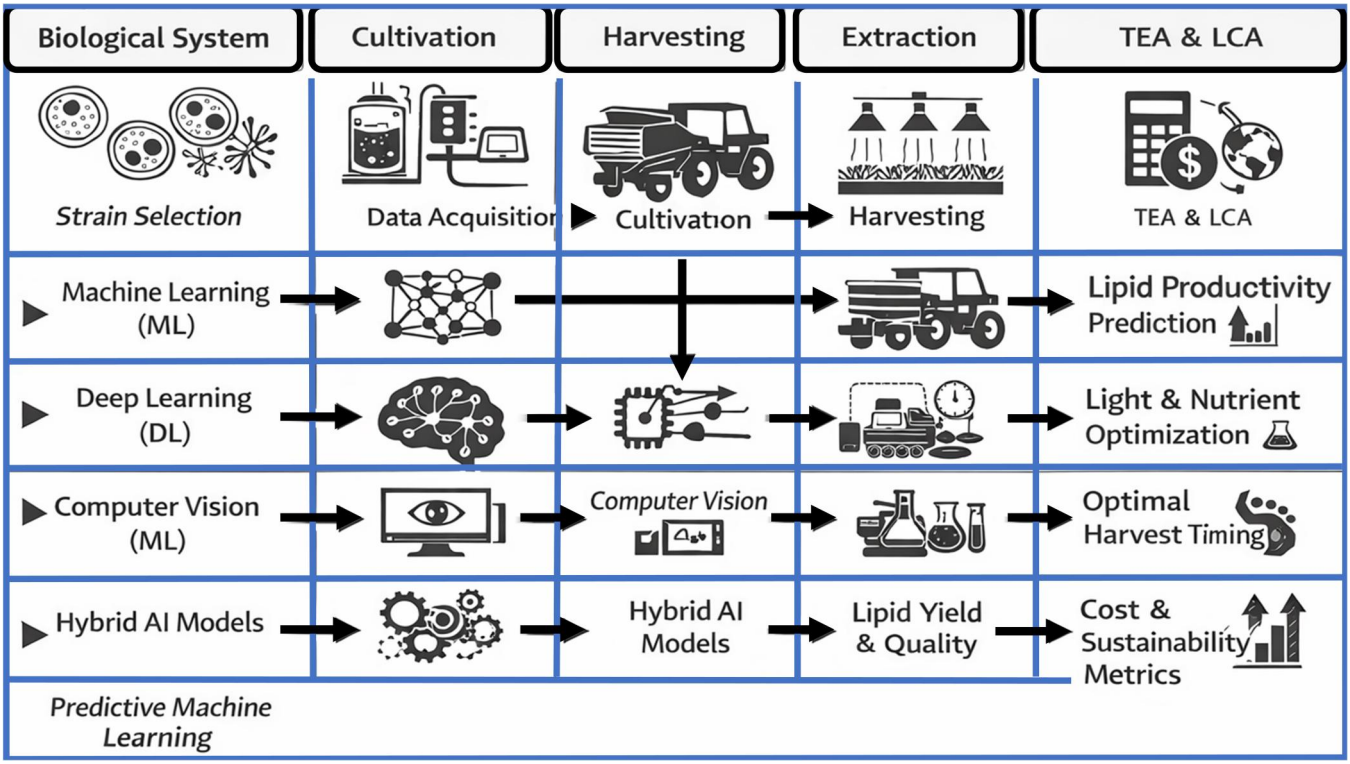


FIGURE 1 | AI techniques mapped to Algal Biodiesel Production.

On the other hand, the hybrid approach entails combining fundamental kinetics with AI models, and this can be generalized as follows:

$$Y = \alpha Y_{\text{mechanistic}} + (1 - \alpha) Y_{\text{AI}},$$

where $Y_{\text{mechanistic}}$ refers to output values obtained from fundamental kinetics or metabolic models, while Y_{AI} refers to the output from the AI model.

Current advancements in hybrid modeling, however, go well beyond straightforwardly combining mechanistic and AI estimates using weighting schemes. Hybrid models that incorporate a mechanistic growth kinetics model and ML correction factor take the form:

$$dX/dt = \mu(S, I, T, \text{pH})\text{X} + \text{fNN}(X, S, I, T, \text{pH}, \theta),$$

where X is the biomass concentration, μ is the mechanistic growth function depending on substrate concentration (S), irradiance (I), temperature (T), and pH level, and fNN is the neural network correction factor expressed as a function of θ. This formulation allows ML models to incorporate biological features that cannot be captured by first-principles modeling, while maintaining the interpretability of mechanistic models. These mathematical models enable accurate analysis of the model dynamics, highlighting disparities between the two approaches. A hybrid modeling technique that combines fundamental principles of chemical reaction kinetics with ML has proven highly effective in bioprocessing operations (Aderemi et al. 2021; Patel 2025).

3 | Application of AI in Algal Biodiesel Production

3.1 | AI-Guided Strain Selection and Metabolic Optimization

Strain selection is currently one of the most critical steps in algal biodiesel production, as it directly affects lipid yield, stress resistance, and process stability. Common methods in strain selection include laboratory screening, a laborious process of testing prospective strains in a controlled setting, followed by strain optimization under nutrient or stress conditions. Although useful, strain screening methods are unsuitable for large-scale use because they often yield inaccurate predictions of how a strain would perform in a process setting. AI methods are changing this trend (Tang et al. 2026).

ML algorithms are increasingly used to predict algal growth and lipid production based on physiological properties, growth conditions, and environmental factors. By combining experimental results from several strains, supervised ML techniques can reveal interesting, yet non-intuitive, relationships among genotype, growth, and lipid productivity that are difficult to establish through empirical approaches alone. This also makes it possible to prioritize strains whose optimal balance of growth rate and lipid content yields favorable properties. Integration of omics data has expanded the potential for AI-assisted strain optimization (Butean et al. 2025; Khan et al. 2025). High-resolution images of how cells react to nutrient stressors, light conditions, and carbon availability are provided by transcriptomic, proteomic, and metabolomic datasets. The identification of metabolic signatures linked to increased lipid biosynthesis or stress resilience has been enabled by AI techniques such as feature

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

selection, clustering, and network inference. Such insights enable rational, priority-based strain-improvement strategies, including adaptive laboratory evolution and targeted genetic modification, while minimizing trial-and-error experimentation (Molla and Meseret 2025). AI also enables comparison of a range of algal taxa to identify understudied taxa with the best biofuel qualities. Unsupervised learning techniques can identify functional similarities that cut across phylogenetic boundaries by classifying strains based on metabolic behavior rather than taxonomy (Venkatrayalu et al. 2026). Given the vast, little-studied diversity of microalgae, which is crucial for developing appropriate biofuels, the capacity to do so is essential. Beyond identifying improved strains, AI also allows dynamic metabolic optimization during culturing. Predictive models can help determine the optimal timing for lipid induction, harvesting, or nutrient modulation, and predict metabolic changes that occur under environmental stress. These methods, when combined with real-time monitoring, enable adaptive control strategies that better balance lipid synthesis and biomass accumulation than static cultivation protocols (Budi Susilo et al. 2025). The generalizability of AI-guided strain optimization across scales and cultivation systems, however, continues to present certain obstacles to its industrialization. This is typically caused by experimental bias and data heterogeneity (Madika et al. 2025). In addition, there is a requirement to connect AI-generated predictions with mechanistic knowledge to facilitate regulatory approval and long-term process stability. For the general development of algal strains to move from an empirical area of biotechnology to a predictive, scalable one, these challenges will be of utmost importance. A summary of AI approaches applied across the different stages of algal biodiesel production, including strain selection, cultivation, harvesting, downstream processing, and techno-economic and life-cycle assessment, is presented in Table 1.

3.2 | Smart Cultivation Systems: Growth Optimization by AI

Optimizing algal growth on an industrial scale is a multifactorial challenge since light intensity, nutrient availability, temperature, pH, mixing, and CO_2 delivery are interconnected. Traditional experimental design relies on either one-factor-at-a-time or limited factorial approaches that cannot capture the nonlinear, multivariate dynamics of algal cultures. An emerging transformative solution is AI-powered smart cultivation systems that enable data-informed optimization of growth conditions and efficient resource utilization in real time (Rayamajhi et al. 2025; Obidi and Bayless 2025).

ML models can combine heterogeneous data, including sensor outputs, environmental measurements, and historical cultivation records, to predict growth trajectories and lipid productivity under variable conditions. Some supervised learning examples include RFs and gradient boosting for accurate prediction of biomass accumulation and for the identification of optimal nutrient dosing schedules. These models minimize costly empirical testing while improving predictive reliability across diverse strains and cultivation systems (Vaughan et al. 2022). Convolutional and recurrent neural networks make image-based monitoring of algal cultures easier. The implementation of AI techniques in high-resolution imaging enables automated analysis of algal health based on density, morphology, and overall health. In contrast, predictive modeling helps forecast stress conditions such as photoinhibition, nutrient deficiency, or contamination (Guo et al. 2025; Sheik, Kumar, Patnaik, et al. 2024; Privat et al. 2025; Sheik, Kumar, Ansari, et al. 2024). One area with great promise is the use of digital twins in algal bioreactors. Digital twins are computer models of physical systems that are constantly updated with real-time information. When digital twins are used with reinforcement algorithms, the technology can make decisions autonomously, exploring thousands of possible ways to do something before execution. This is far faster and cheaper than the physical approach, and this plays an important role where full production capacity needs to be considered (Qian et al. 2024).

Hybrid models that integrate AI with mechanistic models add greater robustness and predictive power by combining process engineering knowledge with data-derived insights. These models could model light distribution, nutrient gradients, and hydrodynamics in bioreactors, thereby providing process-engineering-based explanations that enhance the interpretability and robustness of AI-predicted results (Sierra-Espinosa et al. 2025; Shokravi and Mehravar 2025). This is particularly important for upscaling laboratory results to commercial photobioreactors or open-pond setups, where uncertainties are multiplied. However, to successfully deploy AI-based bioreactor culturing systems, high-quality data, suitable sensors, and efficient communication networks are also necessary. Drift in sensor signals, insufficient data, and issues with real-time computations are among the factors that substantially hinder the development of fully automatic bioreactor culturing systems. However, the integration of AI, digital twins, and smart sensors is revolutionizing algal bioreactors, moving them from reaction-based systems to automated optimizers to effectively bridge the gap between lab results and reality (Sharma et al. 2024; Subburaj et al. 2025).

ML models have been widely used to optimize growth conditions for light intensity, temperature, nutrient concentration, and CO_2 concentration, thereby increasing biomass yield and

TABLE 1 | AI application in algal biodiesel.

Stage	AI approach	Purpose	Reference
Strain selection	AL, DL	Predict lipid yield and growth	Yang et al. (2024)
Cultivation	DL, RL, Hybrid models	Optimize light, temperature, and nutrients	Wu et al. (2025)
Harvesting	Image analysis, ML	Determine optimal harvest time	Imamoglu (2024)
Extraction and conversion	Hybrid modeling, ML	Maximize lipid yield and quality	Geng et al. (2025)
TEA and LCA	ML and Predictive Modeling	Economic and sustainability optimization	Mafat et al. (2024)

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

lipid accumulation (Yeh et al. 2023). This method enables efficient exploration of high-dimensional parameter space, which is quite difficult using traditional methods.

3.3 | Algal Harvesting, Lipid Extraction, and Biodiesel Conversion

Downstream processing accounts for a large share of the costs and energy intensity of algal biodiesel production. This process has been identified as key to the economic viability of algal biodiesel, but current practices are energy- or labor-intensive. This challenge can now be addressed through the use of AI, which can overcome the significant gap between laboratory-scale efficiency and the industrial-level viability required. AI-based harvesting approaches use predictive modeling to optimize timing and technique selection. Supervised ML algorithms can assess growth curves, lipid production trends, and other environmental factors to predict the optimal harvest time that maximizes lipid production while minimizing biomass loss. DL-based image analysis enables real-time monitoring of culture density, shape, and contamination, enabling dynamic adaptation of flocculation, centrifugation, and filtration schemes accordingly (Tummawai et al. 2024).

AI-driven optimization is equally applicable to both lipid extraction and conversion. A variety of ML models have been used to evaluate green extraction techniques, such as supercritical CO_2, ionic liquids, and enzyme-assisted extraction, by identifying extraction efficiency based on algal strain characteristics, solvent properties, and operational parameters (Xie and Zhao 2025; Sarkar et al. 2025). AI frameworks can further optimize transesterification reactions by adjusting catalysts, temperature, and reaction time to maximize biodiesel yield and quality and minimize side products (Yusuff et al. 2023). Hybrid modeling approaches improve process scalability and reliability by fusing data-driven AI predictions. To minimize material and energy waste and accelerate the shift from pilot to industrial scale, these models can replicate downstream processing scenarios that engineers may have before experimental implementation. Furthermore, consistent fatty acid methyl ester profiles required for adherence to biodiesel standards will be supported by AI-based optimization (Hosseini et al. 2025). The end-to-end integration of AI into the downstream process enables a closed-loop, adaptive biological process that leverages data from the growth, harvest, and extraction phases to optimize process control. Not only will these applications increase the efficiency of algal biodiesel production and minimize production costs, but they will also improve the ecological profile of the process. However, issues remain regarding the adoption of integrated datasets and automated platforms that leverage AI for downstream optimization (Maneedaeng et al. 2025).

3.4 | Quantitative Syntheses and Comparative Analysis of AI Models

To provide an objective basis for comparing previous research, a quantitative synthesis of the AI models used in microalgal biodiesel production is presented in Table 2. The synthesis covers information from various processes in biodiesel production, including biomass growth, lipid accumulation, culture optimization, and post-production treatment. Such quantitative comparisons are necessary to advance from mere descriptions of previous studies to evidence-based model selection.

As shown in Table 2, ANNs continue to dominate the field in terms of the number of applications. The popularity of ANNs may be explained by their ability to approximate highly nonlinear interactions between environmental parameters (e.g., illumination intensity, temperature, carbon dioxide concentration, and nutrient levels) and biological phenomena (e.g., biomass and lipid production). Previous research has reported excellent predictive capabilities (R^2 values above 0.90) for ANNs, especially in controlled experimental settings. However, ANN models are generally limited by poor interpretability and overfitting when data are scarce or erroneous (Kumar et al. 2024). On the other hand, ML algorithms like RFs have proven more robust to noise and multicollinearity, and less biased in their predictions, than ANN models. Thus, they can provide useful insights into the key process parameters, thus increasing model interpretability. Moreover, although they tend to have slightly lower predictive accuracy than optimized ANN models in some situations, RF models are better at generalizing across different experimental conditions (Abba et al. 2026). Another recent trend in AI applications to algae involves the use of DL algorithms, such as CNNs. CNN models work best with image data and allow researchers to automate the analysis of cell morphology, growth kinetics, and contamination identification from images produced by microscopes or sensing devices. In this way, they reduce the need for time-consuming observation and enable high-throughput phenotyping of algal strains. Still, their performance strongly depends on the amount and quality of available annotated data (Sultana et al. 2022). Recent advances in modeling techniques have sparked growing interest in hybrid models that combine mechanistic and ML approaches. Hybrid models are more efficient than other models because they combine the strengths of mechanistic models for explaining underlying processes while still providing accurate predictions of outcomes. Hybrid models can achieve prediction errors of less than 5% compared to other models. The generalization ability of hybrid models makes them more useful in scaling up production processes. An important development arising from the synthesis of information is the gradual shift from purely data-driven models to hybrid intelligence that combines domain expertise with data-driven learning (Rangel Pinto et al. 2024).

However, some issues persist despite these improvements. Comparability of model performance remains problematic due to inconsistent datasets, metrics, and methodologies. Moreover, some models were not validated beyond a laboratory scale, and there is insufficient evidence of their scalability and robustness at a pilot or commercial level. All these points underscore the need for standardized approaches to testing and benchmarking AI models, as well as for building large-scale datasets for their training. In general, quantitative analysis shows that although AI models have shown great promise, further development will require hybrid modeling approaches, improvements in data infrastructure, and comprehensive validation (Kushwaha et al. 2023; Malashin et al. 2026). As illustrated in Table 2, both ANNs and ensemble learning algorithms have achieved high

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Comparative study of different artificial intelligence models employed in biodiesel production from microalgae, focusing on types of models, application phases, inputs used, performance measures, and important results obtained.

S. No	AI model	Category	Application stage	Input parameters	Output variable	Dataset type	Performance	Key findings	Reference
1.	ANN	Data driven	Biomass prediction	Light, temperature, CO_2, nutrients	Biomass concentration	Experimental	R2 = 0.95 RMSE = 0.12	Accurate nonlinear growth prediction	Kumar et al. (2024)
2.	CNN	Deep learning	Growth monitoring	Microscopic images	Growth rate	Image dataset	Accuracy = 90%	Effective image-based monitoring	Sultana et al. (2022)
3.	Random forest	Ensemble ML	Lipid accumulation	Nitrogen, temperature, Salinity	Lipid content (%)	Experimental	Accuracy = 92%, MAE = 3.5	Robust to noisy and high-dimensional data	Abba et al. (2026)
4.	SVM	Machine learning	Biomass prediction	pH, C2, temperature	Biomass yield	Experimental	R2 = 0.89	Good generalization with limited data	Meenatchisundaram et al. (2024)
5.	LSTM	Deep earning	Time series growth prediction	Temporal environmental data	Biomass over time	Time series	R2 = 0.91	Captures temperature dynamics effectively	Syed et al. (2024)
6.	Gradient boosting	Ensemble ML	Cultivation optimization	Multi-factor environmental variables	Biomass productivity	Experimental	R2 = 0.93	Improved prediction over F in some cases	Oruganti et al. (2023)
7.	ANN + Kinetic model	Hybrid	Cultivation modeling	Environmental = Kinetic inputs	Biomass = Lipid	Mixed	RMSE < 5%	High accuracy interpretability	Rangel Pinto et al. (2024)
8.	ANN + Genetic algorithm	Hybrid	Process optimization	Nutrient, light, temperature	Lipid yield	Experimental	Error < 7%	Optimized cultivation parameters	Kushwaha et al. (2023)
9.	Decision tree	ML	Harvesting efficiency	Flocculant dosage, pH	Recovery efficiency	Experimental	Accuracy = 85%	Simple and interpretable	Wu et al. (2023)
10.	CNN + ANN	Hybrid	Image+ growth modeling	Images + environmental data	Biomass growth	Mixed	Accuracy = 93%	Multi-model modeling advantage	Xu et al. (2025)
11.	Deep neural network	Deep learning	Lipid Prediction	Multi-variable inputs	Lipid content	Experiment	R2 = 0.4	Handle complex interaction	Wu et al. (2025)
12.	Reinforcement learning	AI optimization	Process control	Real-time sensor data	Optimized yield	Real time	Yield increased by 20%	Adaptive system control	Sirohi et al. (2026)

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

prediction accuracy, with R^2 values exceeding 0.90 in many studies. Moreover, hybrid modeling techniques have proven to be more generalized and accurate than data-driven models alone (Pavendan and Nagarajan 2022; Araújo et al. 2026).

4 | AI-Powered Techno-Economic and LCA

Economic viability and environmental sustainability remain challenges for algal-based biodiesel production. Although AI-assisted strain identification, optimization, harvesting, and lipid extraction enhance efficiency, a holistic process analysis integrating these AI technologies is a crucial step toward converting a "success in a lab" process into a practical "success at a plant" process. TEA studies, along with LCA analyses, serve as research frameworks for understanding capital costs, operating costs, energy balance, GHG emissions, and related metrics. Performed with AI assistance, these studies turn out "dynamic, predictive, and adaptive," thereby optimizing a complete "biofuel production value chain" (Zahra et al. 2025; Giacomella 2021). ML algorithms can assimilate data from cultivation, harvesting, and conversion processes to estimate overall production costs, energy efficiency, and return on investment. An AI model simulation of thousands of scenarios can identify the cost-optimal solution that accounts for variations in productivity rates, climate patterns, and market trends (Zaman et al. 2025). Reinforcement learning algorithms and optimization processes can generate a sequence of solutions that reduce overall energy usage and maximize lipid production. Similarly, AI-assisted life-cycle assessment enables real-time analysis of environmental impacts, such as carbon footprint, water use, and chemical consumption (Xian et al. 2025). Predictive models that consider cultivation parameters, bioreactors, processing, and analysis across scenarios can support sustainable decision-making based on these assessments. Even with the identification of "hot spots" in a bioprocess for high energy input or waste, AI-assisted measures such as incorporating waste valorization or optimizing bioreactor illumination can help reduce environmental impact while remaining economically viable (Satish Kumar et al. 2022). Hybrids integrating mechanistic process models with AI-driven predictions can increase certainty and interpretability, making the models more acceptable for regulatory and stakeholder purposes. Such models can not only analyze existing processes but also model performance at scale or with new feedstocks, thereby providing a guide for the future. Integration of AI with TEA and LCA establishes a closed-loop decision-support system in which operational data continuously feed the predictive model, helping optimize operating parameters. Thus, the application of AI helps bridge the gap between algal biorefineries with experimental setups and scaled-up facilities capable of effectively producing eco-friendly biofuels in an economically viable manner, thereby promoting green chemistry-driven practices (Lin et al. 2025). The economics of developing microalgal biodiesel are among the most important factors. The reported minimum biodiesel selling price (MBSP) ranges from 4 to 10 USD per liter, which is considerably higher than that of fossil diesel (around 0.8 to 1.2 USD per liter, He et al. 2025). Cost estimation studies show that the cultivation and harvesting phases account for over 60% of total operational costs due to high energy inputs and low biomass production rates (Slade and Bauen 2013).

AI-based optimization methods have proven their worth in improving the economics of microalgal biodiesel production by increasing efficiency. This includes increasing biomass productivity by 10%–30% by optimizing environmental conditions, decreasing energy consumption during harvesting operations by 15%–25% through predictive flocculation, and increasing lipid yield to around 20%. The sensitivity analysis shows that the growth rate, lipid content, and energy consumption have the greatest impact on costs (Yang et al. 2024; Onay and Onay 2026; Tripathi et al. 2024). LCA is a holistic analysis of the environmental sustainability of algae biodiesel systems. The reported energy ratios generally range from 1.5 to 3.0, indicating a favorable energy output ratio under optimal conditions. Emissions of greenhouse gases (GHGs) are projected to be around 20–60 g CO_2-equivalent per MJ, considerably less than fossil fuel-based diesel (~85 g CO_2-eq/MJ) (Costantini 2025; Semwal and Joshi 2026). However, despite these benefits, a few environmental issues remain unresolved. Excessive water use (between 3000 and 5000 liters per kg of algae biomass) and high nutrient requirements (particularly nitrogen and phosphorus) raise sustainability concerns (Yang et al. 2011). The application of AI can help address these environmental issues by enabling more efficient use of resources and reducing waste. Combining AI with life-cycle assessments enables continuous monitoring and optimization of environmental impacts. According to techno-economic studies, the cost of producing algal biodiesel remains relatively high compared to traditional fossil fuels due to the energy required for production (Gu et al. 2020). Nevertheless, using AI-based optimization methods may help reduce production costs by improving management and resource utilization (Sampaio et al. 2024). Despite the promising prospects for cost savings enabled by AI, AI alone will not be enough to bridge the economic disparity between the production costs of algal biodiesel and fossil diesel. Estimates indicate that the cost of producing algal biodiesel is around 4–10 USD per liter, compared to fossil diesel at around 0.8–1.2 USD per liter. While AI-powered optimization is expected to reduce costs through increased efficiency in algae cultivation and harvesting, lower energy requirements during harvesting, and higher lipid productivity, these savings alone cannot ensure cost parity. Thus, it appears that economic feasibility will depend on its integration with wastewater processing, CO_2 capture utilization, and valuable coproducts (Maroušek et al. 2023; Yaashikaa et al. 2022).

5 | Bridging the Commercialization Gap in Algal Biodiesel Systems

Although extensive research has been conducted on the production of biodiesel from microalgae and many new technologies have been developed over the last few decades, its commercialization remains limited. The initial hopes for the economic viability of algae biofuel technology have not yet materialized, largely due to ongoing challenges with cost, scalability, and cultivation. According to current estimates, algal biofuel production costs much more than conventional fossil-fuel production. Among the main problems in the commercialization of algae biodiesel are issues related to scaling up. While small lab-scale studies yield positive results, they cannot be easily scaled up to pilot- and industrial-scale facilities. Factors such as light availability and distribution, mass-transfer

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

issues, contamination risks, and variability in the external environment pose significant concerns (Tripathi et al. 2025). AI is a useful technology that can help solve these problems in predictive modeling, process optimization, and real-time control. ML models could help optimize growth, predict lipid formation, and improve harvesting. However, there are certain limitations to the actual implementation of AI technology in industrial processes. For example, one major limitation is the lack of access to sufficient, high-quality data for training ML models (Mir et al. 2025). The second major limitation of such models is insufficient validation across different scales. AI models are usually tested under laboratory conditions, without demonstrating their efficiency at pilot or industrial scales. This means that there is still a considerable distance between theoretical optimization and real-world application. Apart from that, volatile fuel prices and the lack of incentive policies are also factors that complicate the implementation of AI in this sector (Coşgun et al. 2023).

According to a review of the existing literature, AI technologies should be considered complementary to other approaches used in biofuel production. For algal biodiesel to become commercially viable, its development and optimization should take into account not only recent advances in AI but also progress in bioprocess engineering, photobioreactors, and genetics. The creation of relevant infrastructure and datasets might play a significant role in bridging the gap between research and industry. However, despite significant advances in research, large-scale commercialization of algal biodiesel faces challenges, including high production costs and the need to scale up the technology. The current literature underscores the importance of an integrative approach to solve the identified problems (Adewuyi 2022; Deepankumar and Kl 2026).

5.1 | Model Transferability and Scalability in AI-Driven Algal Refineries

The most challenging issue in applying AI to algal biodiesel synthesis is the lack of transferability of AI models developed on laboratory data to industrial cultivation systems. Indeed, ML and DL algorithms usually use datasets collected in laboratories with precisely controlled environmental factors, including temperature, light intensity, pH, nutrient levels, agitation rate, and CO_2 availability. In such cases, the biological variance can be reduced, enabling accurate predictions of biomass yield, lipid formation, and process optimization using AI models. Still, the accuracy of such models tends to drop dramatically when moving from the lab stage to either pilot or commercial algal cultivation systems due to increased environmental complexity and variability (Khan et al. 2025).

Unlike laboratory conditions, industrial-scale algal cultivation, especially in raceway ponds and large photobioreactors, involves numerous uncertainties regarding environmental factors. Variations in solar radiation, seasonal temperature changes, precipitation, humidity, wind velocity, evaporation, and the presence of competing microorganisms in the system can strongly affect algal productivity. Furthermore, industrial cultivation systems exhibit heterogeneity at various scales; thus, differences in nutrient concentrations, O_2 levels, CO_2 availability, and light intensity can lead to the formation of microenvironments. The presence of these variables results in biological responses that are not typically included in laboratory datasets, thereby complicating the use of AI algorithms designed on the assumption of predictable system behavior. Domain shift is a well-known computational problem in which there is an inconsistency between the statistical characteristics of the data used to train the algorithm and those in the target environment's dataset. This situation can be caused by fluctuations in environmental parameters, cultivation strategies, reactor design, algal strain selection, or even measurement technology. Therefore, a model exhibiting perfect predictive performance in the lab can face serious problems predicting outcomes in an actual industrial setting due to poor generalization. Overfitting exacerbates this problem when insufficient data are used during model training, due to the high variability of biological responses (Yahaya et al. 2025; Udaypal et al. 2026).

Another limitation is the failure of strictly data-driven models to predict system performance under operational conditions not captured in the training data sets. Biological systems operate in a highly complex environment characterized by nonlinear relationships among metabolism, uptake processes, stress, and environmental interactions. Prediction accuracy may be seriously compromised when cultivation is conducted under conditions the model has not seen before. This limitation is more critical during scaling up, where the major issues include hydrodynamics, mass transfer, self-shading, biofilms, and contamination. Traditional ML algorithms often fail to understand these principles and hence may be limited in their predictive power. To circumvent these drawbacks, there is an increasing focus on developing a hybrid modeling framework that integrates AI algorithms with process-based mechanistic models. This class of models uses biological kinetics, mass balances, and metabolic constraints, along with engineering principles and ML methodologies, to enhance predictive power and enable extrapolation beyond their training datasets. Some of the novel strategies being adopted include PINNs, Neural ODEs, and mechanistic ML, in which biological understanding can be used to constrain the learning process, thereby reducing model uncertainties (Hosseini et al. 2025; Ahmadi et al. 2026). Moreover, the development and deployment of AI in biodiesel production can benefit from additional methods like transfer learning, domain adaptation, and digital twins. Transfer learning is possible thanks to models trained on vast datasets collected during laboratory experiments, which can then be fine-tuned on relatively small datasets collected in industrial environments, thereby sparing the costs of massive experimental campaigns. Domain adaptation is another method that helps reduce discrepancies between training and testing environments, enhancing a model's robustness under varying conditions (Yang et al. 2026). Also, the use of digital twins, which are virtual replicas of physical cultivation systems continuously fed with sensory information, represents another important avenue for the deployment of AI in the algae biofuel industry. When coupled with reinforcement learning algorithms, digital twins can be used to test multiple scenarios and select the optimal one for cultivation (Haq et al. 2026). However, the successful deployment of AI in industry still requires efforts to develop standardized protocols for data collection, high-quality, multi-scale datasets, sensor networks, and proper validation

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

procedures across laboratory, pilot, and full-scale facilities. In addition, it would be necessary to focus future research on developing generalizable, robust, and interpretable AI models that can operate under highly variable cultivation conditions.

6 | Integrated Analytical Framework for AI-Driven Algal Bioprocess Optimization

With this in mind, a systematic methodology for analyzing and comparing modeling techniques in AI is introduced to evaluate and compare methods used in algal biodiesel production. In this regard, the following AI-based models can be categorized into three groups according to their main characteristics: Data-driven models, such as ANNs, RF, SVMs, and DL models, like CNNs and long short-term memory (LSTM) networks. These models are based solely on data and are highly efficient at predicting nonlinear interactions among input variables. Nevertheless, interpretability is an issue, and large amounts of data are required for training (Singh and Sharma 2025). Models are founded on first principles, such as kinetics, metabolic pathways, and mass balance equations. These models have high interpretability and a theoretical basis, but they are prone to limitations, such as simplifications in solving the governing equations and limited ability to capture biological variability. Models that combine data-driven and mechanistic approaches can overcome their limitations, thereby increasing prediction efficiency, robustness, and accuracy. Objective comparison is made possible by evaluating each modeling method using specific criteria, such as: Accuracy of prediction (e.g., R^2, RMSE), Complexity and computational demands, Interpretability and transparency, Scalability from lab scale to pilot and industrial scale, Necessities and sensitivities to data. Additionally, the proposed framework considers the applicability of modeling methods at various stages of the process (e.g., the cultivation phase or downstream processing). For example, data-driven models are better suited to online measurement, while hybrid models are ideal for process optimization. It should be noted that this framework can serve as a basis for developing standardized benchmarking procedures, which are currently lacking in the field. Hybrid models that combine knowledge-based and data-driven techniques have shown promise in bioprocessing systems, offering the potential to improve model reliability and scalability. Hybrid models facilitate comparisons of different modeling techniques in terms of efficiency, interpretability, and data requirements (Ansari et al. 2026).

7 | Implementation and Deployment Strategies for AI-Enabled Algal Biorefineries

The realization of AI in algal biodiesel processes requires a three-step approach: data collection, model training, and algorithm implementation. These steps can be represented by a pipeline incorporating experimental and computational elements. The first step in this pipeline is data collection. Data are required from different sources, including laboratory studies, pilot processes, environmental sensing devices, and imaging apparatuses. Important variables here include light intensity, temperature, pH value, nutrient concentrations, CO_2, and biomass density. The use of Internet of Things (IoT) sensors contributes to continuous data monitoring. The second stage includes data preprocessing. In this stage, the raw data have to be properly filtered, processed, and formatted for further analysis. Specifically, this includes noise reduction, the elimination of irrelevant information, and the treatment of missing values and normalization procedures. Stage three involves selecting and developing an appropriate AI-based model for the specific problem being solved, such as regression, classification, or time-series prediction. For example, ANN and RF models are widely used in predictive analytics, while the LSTM neural network is a powerful tool for time-dependent processes. In cases where mechanistic insights are required, a hybrid modeling approach may be implemented (Malik et al. 2026). Model training and validation procedures constitute stage four of AI-based analysis. Model accuracy is typically measured using R^2, root-mean-square error (RMSE), and mean absolute error (MAE). At the final stage, ML-trained models are deployed in process control systems for ongoing monitoring and optimization of culture parameters. Examples of such tasks are the automatic tuning of cultivation conditions and the maintenance of equipment in the best possible shape. Common software and hardware tools for carrying out the workflow described above include Python libraries such as TensorFlow, PyTorch, and Scikit-learn, as well as visualization tools such as MATLAB and R. Cloud computing infrastructure and edge computing devices are highly beneficial for deploying AI-based solutions. These issues must be addressed for AI-based algal biorefineries to be implemented on an industrial scale. The use of AI systems in algal biorefineries is also increasingly becoming a reality, thanks to digital technology developments such as the IoT sensors and cloud computing data platforms (Saravanakumar et al. 2024; Park et al. 2024).

8 | Challenges and Limitations

Although applying AI to algal biodiesel has the potential for revolutionary change, the challenges of implementation limit its effectiveness and predictive power. High-quality, high-resolution, multi-scale genomic, transcriptomic, metabolomic, and bioprocess-related data are needed for AI models to operate effectively. However, in algal bioprocesses, such high-quality data can be sparse, noisy, or heterogeneous due to differences in strains, operational conditions, or analytical methods. Inadequate or unbalanced datasets can result in overfitting, poor generalization, and erroneous predictions, thereby reducing the accuracy of AI-assisted decision-making (Onianwah et al. 2025). Advanced DL models, although efficient, tend to be "black boxes" in the sense that, apart from being highly efficient, little mechanistic insight is obtained. In biotechnology, especially in the regulated, industrial biorefineries sector, it is important to understand the rationale behind the AI tool's recommendation (Ge et al. 2025). For example, models developed from laboratory- or pilot-scale data may not readily adapt to full-scale reactor environments, where the effects of environmental inhomogeneities, hydrodynamics, and operational differences are more pronounced. Addressing the size disparity calls for innovative solutions such as cross-validation, domain adaptation, or mechanistic models (Arulrajah et al. 2025). For AI to be effectively implemented in real-time scenarios, high-quality sensing infrastructure is essential. In today's industrial environment, such infrastructure might be either rudimentary or economically

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

infeasible. Second, data management for sensor networks is a concern in its own right, while cybersecurity is another issue (Abdullah et al. 2025; Dubey and Dutta 2025). The enhanced autonomy in AI-based bioprocesses also raises questions about accountability, reproducibility, and the ethical controls that should be exercised. The regulatory environment on the use of AI-based biotechnologies is still under development. The energy-intensive biocomputation may pose environmental problems if the algorithms are not optimized for sustainability (Mirakhori and Niazi 2025). The successful application requires collaboration across fields such as biotechnology, process engineering, data science, and modeling. Knowledge gaps in interdisciplinary education may impede the integration of AI in traditional algal biofuel production.

To overcome the mentioned challenges, collaboration is needed to standardize data capture, design explainable AI models, improve sensor networks, and augment AI models with mechanistic insights (Guo et al. 2025). All these challenges must be addressed if the potential of AI models for large-scale, sustainable algal biodiesel production is to be harnessed. Since biological datasets generated during algal cultivation often contain missing data, noise, batch effects, and inconsistent measurements, biology-oriented solutions are necessary. Missing data can be imputed through Kalman filtering techniques. Batch effects arising from multiple experimental batches can be normalized using methods such as ComBat. Transfer learning leverages prior knowledge from well-researched strains to make predictions for other, less well-known algae. Robust Bayesian methods could reduce the impact of biological outliers on predictions, while adaptive calibration algorithms could help mitigate sensor drift. The combination of all these techniques will greatly enhance the performance of prediction models in algal biotechnology (Metodiev et al. 2026).

The quality of sensor data significantly influences the performance of AI-based monitoring and control. The sensors operate under harsh conditions such as biofouling, fluctuating salt concentrations, microbial contamination, sedimentation, and stress in industrial microalgae cultivation processes. All these aspects influence sensor performance and introduce signal noise and calibration drift that affect the precision of process predictions and decisions made by AI models. Since AI-based systems require continuous data to operate, sensor errors can lead to inaccurate biomass estimates and incorrect nutrient dosing during cultivation. Therefore, sensor reliability is a key drawback of autonomous algal biorefineries. To address these issues, future AI-based systems will use a sensor redundancy approach, in which multiple sensors collect data on the same process parameter. There are also methods of data fusion that combine information from several sensors to produce more reliable estimates of process parameters. Another way to reduce the importance of physical sensors is to implement virtual (soft) sensors based on ML algorithms. Predictive maintenance algorithms are also among the innovations developed. Using analysis of sensor performance trends, the AI algorithms will predict sensor malfunctions, enabling timely maintenance without operational disruptions. The adoption of robust sensor networks, soft sensing techniques, a data fusion approach, and predictive maintenance is crucial for achieving high-quality data and ensuring the reliability of the AI monitoring system in large algal biorefinery projects (Ali et al. 2025; Karthikeyan et al. 2026).

Some key challenges and possible strategies for implementing AI in algal biodiesel production are highlighted in Table 3.

9 | Future Outlook: Toward Autonomous Algal Biorefineries

The planned integration of AI into algal biodiesel production is expected to shift bioprocessing from human-centered, manual, traditional approaches to autonomous, self-optimizing algae biorefineries. In an algae biorefinery, AI will function not only as an analytical aid but, more importantly, as a key orchestrator.

9.1 | Autonomous Cultivation and Digital Twins

The next-generation biorefineries will exploit enhanced digital twin technologies—digital replicas of biological and engineering systems that assimilate sensor data and use predictive models in real time. Moreover, digital twins using reinforcement learning algorithms will enable autonomous optimization of operating procedures by adjusting variables such as illumination levels, nutrient delivery, and mixing. Self-optimized processes maximize efficiency and lipid yield while maintaining microbial cultures (Sheik, Kumar, Ansari, et al. 2024; Subburaj et al. 2025).

9.2 | Integration of Multi-Omics Analysis With AI-Based Metabolic Engineering

Indeed, future biorefineries designed with AI will incorporate genomic, transcriptomic, and metabolomic data to optimize metabolism through rational metabolic engineering. Furthermore, through a continuous learning process informed by real-time cultivation results, the AI will provide appropriate interventions to increase lipid productivity and robustness to environmental variations (Yadav et al. 2025).

TABLE 3 | Challenges and limitations of AI in algal biodiesel (Omole et al., 2024; Jathar et al. 2025; Pusuluru et al. 2026).

S. No.	Challenge	Impact	Mitigation strategies
1.	Data quality and heterogeneity	Reduced model accuracy	Standardized, cross-validation
2.	Model interpretability	Regulatory and operational trust	Explainable AI, mechanistic integration
3.	Scalability	Lab to industrial transfer	Digital twins, mechanistic integration
4.	Sensor and infrastructure limitations	Real-time control difficulty	Advanced sensor network, IoT integration

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

9.3 | Smart Downstream Processing

Autonomous biorefineries will involve AI-based harvesting, extraction, and conversion, resulting in closed-loop systems that provide optimized efficiency, resource use, and product yield. AI-based ML will aid in optimizing harvest times, extraction procedures, and transesterification, thereby reducing energy and reagent or chemical consumption. The union of AI technologies and green chemistry approaches will therefore ensure that this process remains eco-friendly (Hamood ur Rehman et al. 2025; Jayabal and Prabhakar 2025).

9.4 | Sustainability and Techno-Economic Optimization

With AI support in the assessment framework, there will be the ability to optimize at the system level, with operators anticipating potential bottlenecks, so that economic outcomes can be predicted and even environmental impacts can be mitigated through such technology.

9.5 | Outlook and Opportunities

Although there remain technical, infrastructure, and regulatory hurdles to address, based on what is seen in AI for algal biofuels, there is a promising near-term future for self-regulating, predictive, and optimizing biorefineries. They will provide a strong opportunity for algal biodiesel to become economically viable, sustainable, and scalable, thereby driving a major shift in the biofuels sector.

10 | Sustainability and Broader Impact

AI-supported algal biodiesel production systems offer immense potential for the global energy transition toward sustainable sources, providing a low-carbon alternative to fossil fuels. Algae have several intrinsic advantages over other terrestrial biofuel sources, owing to their high photosynthetic rates, rapid growth, and capacity to accumulate lipids in abundance when cultivated under appropriate conditions. In addition, algae do not compete with crops for farmland, as they can be cultivated on non-arable land, such as wastewater farms, and even in saline environments. In terms of environmental impact, algal biodiesel has tremendous potential to reduce GHG emissions throughout the process life cycle. AI further adds value in this respect by optimizing various processes and reducing waste. This includes optimizing energy use in the process, adding nutrients, and minimizing waste. Combining cultivation with carbon dioxide capture techniques, such as flue gas utilization, further enhances the scope for reducing greenhouse gas emissions in industry. However, despite the benefits outlined above, the sustainability of the algae biodiesel system should be assessed holistically at the systems level. Scaling up the system is associated with greater negative environmental consequences, such as high-water consumption, intensive nutrient demand (primarily nitrogen and phosphorus), and risks associated with the cultivation system at an extensive scale (Pasha et al. 2021). AI can help solve these problems, in part, by predicting and optimizing resource utilization and allocation; however, technological development may be insufficient to address these issues. Moreover, the economic factor must be considered, as it is crucial to sustainability. The cost of current biodiesel production is much higher than that of fossil fuels and other biofuels, thereby hampering the technology's competitiveness. Nevertheless, the application of AI can optimize and economize production, making it more cost-effective, provided that technological advancements in this sphere occur. It is essential to note that sustainability cannot be reduced to economic and environmental factors alone. A range of socio-economic factors can be affected by the use of AI technologies in biodiesel production. Additionally, the ethical and governance issues surrounding the deployment of AI technologies, such as data ownership, transparency, and algorithmic accountability, will become increasingly important when these technologies are applied in bioindustrial systems. Ensuring that technology is accessed equitably and preventing negative environmental or societal effects will be key (Deepankumar 2025).

Although the adoption of AI-enhanced algal biodiesel production systems can provide an innovative solution to achieving sustainable, carbon-neutral energy, its realization must be well-balanced and scientifically substantiated. Future studies should develop holistic approaches to sustainable development, enhance resource-use efficiency, and integrate innovations with appropriate policies and regulations. Microalgal biofuels are preferable to traditional biofuels from a sustainability standpoint because of their higher productivity and minimal land-use competition; however, large-scale use of these fuels should account for resource use and environmental impacts. Energy savings enabled by AI optimization are crucial for achieving greater sustainability. Another equally crucial yet frequently disregarded aspect is the environmental impact of AI. The process of training DL models, along with supporting cloud computing systems, may entail significant energy use, leading to an irony in which energy-hungry computational processes are used to design environmentally friendly biofuels. In this context, recent research has indicated that large-scale ML training may consume hundreds to thousands of kilowatt-hours of electricity, depending on the size of the model. Therefore, future AI-enabled biorefineries will have to rely on energy-efficient algorithms, model compression strategies, edge computing designs, and data centers powered by renewable energy sources (Jena and Agarwal 2026).

AI has been highly beneficial for optimizing processes in microalgal biotechnology; however, it does not resolve the fundamental resource limitations of industrial-scale cultivation. Industrial-scale production of microalgae would require large amounts of water, nitrogen, phosphorus, carbon dioxide, and energy inputs. These limitations would be physical and economic in nature, something that AI is not capable of producing or resolving. Challenges such as freshwater limitation, high fertilizer costs, and limited phosphorus availability remain key obstacles to the industrial application of microalgae-based biodiesel systems. Of all the limitations, water and nutrients would be very important. Depending on the type of cultivation system used and climatic conditions, large-scale algal facilities could use substantial amounts of water for evaporation, harvesting, and biomass processing. Nitrogen and phosphorus are also macronutrients necessary for algal growth and lipid formation.

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Excessive reliance on artificial fertilizers is costly and poses sustainability issues, as it competes for resources with agriculture (Sirohi et al. 2026). However, there are many ways AI can improve resource utilization efficiency through predictive modeling, real-time monitoring, and intelligent process control. Algorithms used in ML can make predictions about nutrient needs based on historical data on crop cultivation, environmental factors, and growth kinetics. As a result, nutrients will be supplied in response to cellular needs, thereby avoiding nutrient losses from overfeeding. Thus, this approach reduces fertilizer consumption without decreasing yield. AI can also help optimize water use in cultivation systems. Using sensor networks and predictive analytics, AI can monitor parameters such as biomass concentration, temperature, evaporation rate, oxygen levels, and water salinity. Real-time analysis of all these variables will help detect inefficiencies in water use and take necessary measures to prevent water loss. Predictive controls will also help adjust water replenishment schedules. One more area where AI technology can be applied is water recycling and nutrient recovery. In an industrial setting, a large portion of cultivation media can be recycled after treatment. With the use of an AI-based monitoring system, it will be possible to assess water parameters and determine the optimal treatment procedure (Venkatrayalu et al. 2026). Moreover, ML algorithms can help increase the efficiency of nutrient recovery from wastewater, thereby reducing the need for external fertilizers. The above-mentioned possibilities are beneficial for creating circular cultivation systems with low resource requirements. Another contribution of AI to resource conservation is maximizing biomass production per unit input. Using sophisticated optimization algorithms, it will be possible to determine the most efficient combination of environmental factors such as light intensity, temperature, pH, CO_2 concentration, and nutrient levels. The use of reinforcement learning and digital twins enables adjustment of the cultivation process in response to changing environmental factors while maintaining maximum efficiency. Thus, there will be higher yields of biomass and lipids, along with lower water and nutrient consumption. In addition, AI helps integrate microalgal cultivation with wastewater treatment technology. Predictive models will optimize the efficiency of nutrient recovery, hydraulic retention time, and biomass harvesting. Therefore, wastewater can become an additional source of both water and nutrients (Sharma et al. 2026).

11 | Explainable AI: The "Black Box" Problem in Algal Biotechnology

Despite the remarkable ability of AI algorithms to predict outcomes, their practical application in algal biotechnology is hindered by a core issue in DL models known as the "black-box" problem. In contrast to traditional models based on the principles of biology and physics, deep neural networks use numerous computational steps to detect complex non-linear patterns in data. While these models tend to produce better predictions, their reasoning remains unclear to humans (Haq et al. 2026). Thus, this aspect makes it challenging to implement AI-based technologies to optimize processes, interpret results, comply with regulations, and make informed business decisions. In the biodiesel industry, AI models are utilized to predict biomass productivity, lipid synthesis, nutrient uptake, cultivation, and harvesting efficiency. Nevertheless, plant managers and researchers would find it difficult to trust model advice when unable to comprehend how certain predictions are generated. For instance, AI models could suggest that reducing nitrogen feeding or changing illumination would improve lipid yield; however, without understanding the underlying principle behind such a recommendation, operators may find it hard to evaluate its reliability. As a result, in large-scale commercial settings, the inability to explain AI-based advice may lead to considerable financial losses due to poor management decisions. To address these difficulties, scientists have turned their attention to developing methods for explainable artificial intelligence (XAI). The goal of these tools is to increase transparency by explaining which input variables affect the output parameters (Krejcar et al. 2026). Various methods are used to make biological models more transparent. Feature importance analysis will allow us to identify which variables determine prediction accuracy. At the same time, the Local Interpretable Model-Agnostic Explanations (LIME) tool creates simple local approximations to explain the operation of complex models. Likewise, SHapley Additive exPlanations (SHAP) measure the effect of each variable in a dataset on predictions. SHAP analysis will help identify which factors (temperature, irradiance, nutrients, CO_2 concentrations, and pH) play the greatest role in determining the predicted amounts of biomass and lipids (Aasim et al. 2026).

Moreover, modern DL methods include innovative technologies such as attention mechanisms and saliency maps that enable scientists to visualize the information processing carried out by the network and identify influential features. This approach allows scientists to understand better the biological processes occurring within the ecosystem and to correlate AI results with them. Another promising direction for investigation is hybrid modeling that combines mechanistic and machine-learning models. By integrating biological constraints, biochemical connections, and kinetic information into AI-based systems, this approach could enable us to achieve high predictive accuracy without sacrificing mechanistic interpretability. Still, this remains an open problem, and even the best explainability approach fails to make any AI system fully interpretable or free of its inherent opacity. The explanations provided by XAI techniques can be approximate and uncertain. Therefore, the validation of AI-assisted models on independent datasets, robustness analysis, and experimental validation are indispensable for achieving trust in AI decision support (Inamdar et al. 2026). Further development of interpretable, robust, and biologically grounded AI systems capable of producing both high-quality predictions and credible scientific explanations is vital for future success. This will allow us to use AI as an important decision support tool in our commercial algae biorefineries.

12 | Conclusion

AI is found to be a revolutionary technology for the production of algal biodiesel. AI is used for predictive strain screening, optimal growth techniques, efficient harvesting, and effective downstream processing. AI combines high-dimensional biological data with process knowledge to offer a systems-level approach to overcome trial-and-error methodologies. The convergence of ML, DL, reinforcement learning, and hybrid

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

modeling techniques with algal biotech has demonstrated improved lipid productivity, efficiency, and scalability. Furthermore, by linking lab innovation breakthroughs to the technical scalability of biofuels within the broader context of the circular economy and climate change mitigation, AI-based TEA tools enable decision-making. Despite significant advancements, several issues remain unresolved, including data, interpretation, scaling, and infrastructure. These will be essential for their application in fully autonomous, self-optimizing algal biorefineries. The emphasis for future research should be on developing resilient datasets, interpretable AI, digital twins, and sustainable process design, in which AI is applied for sustainability as well as efficiency and is grounded in the principles of green chemistry. AI has the potential to transform the bioenergy industry, making algal biodiesel production a sustainable, environmentally friendly process that was previously only an interesting laboratory idea.

Author Contributions

Gyanendra Tripathi: conceptualization, methodology, investigation, data curation, writing – original draft.

Funding

The author has nothing to report.

Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

References

Aasim, M., E. S. Aydin, I. Korkut, et al. 2026. "Machine Learning and Design of Experiment for Enhanced Biomass Using N_2 and CO_2-Enriched Biochar at Different Pyrolysis Temperatures." *Biomass and Bioenergy* 208: 108803.

Abba, S. I., S. R. Chowdhury, A. G. Usman, and S. A. Razzak. 2026. "Experimental and Nature-Inspired Modeling of Lipid Accumulation in Microalgal Biomass for Biofuel Production." *Biomass and Bioenergy* 212: 109255.

Abdullah, M., H. A. Malik, A. Ali, et al. 2025. "AI-Driven Algae Biorefineries: A New Era for Sustainable Bioeconomy." *Current Pollution Reports* 11, no. 1: 21.

Aderemi, B. A., T. O. Olwal, J. M. Ndambuki, and S. S. Rwanga. 2021. "A Review of Groundwater Management Models With a Focus on IoT-Based Systems." *Sustainability* 14, no. 1: 148.

Adewuyi, A. 2022. "Production of Biodiesel From Underutilized Algae Oil: Prospects and Current Challenges Encountered in Developing Countries." *Biology* 11, no. 10: 1418.

Ahmadi, N., Q. Cao, J. D. Humphrey, and G. E. Karniadakis. 2026. "Physics-Informed Machine Learning in Biomedical Science and Engineering." *Annual Review of Biomedical Engineering* 28, no. 1: 309–336.

Alber, M., A. Buganza Tepole, W. R. Cannon, et al. 2019. "Integrating Machine Learning and Multiscale Modeling—Perspectives, Challenges, and Opportunities in the Biological, Biomedical, and Behavioral Sciences." *NPJ Digital Medicine* 2, no. 1: 115.

Ali, H., R. Safdar, J. Liu, et al. 2025. "Hybrid Fusion Paradigm in Advanced Process Monitoring: A Panoramic Review and Future Perspectives." *Industrial & Engineering Chemistry Research* 64, no. 47: 22465–22514.

Alzahrani, R. A., M. Aljabri, and R. A. Mustafa Mohammad. 2025. "Ad Click Fraud Detection Using Machine Learning and Deep Learning Algorithms." *IEEE Access* 13: 12746–12763.

Amina Raza, R. 2024. "Nonlinear Dynamics in Complex Systems: A Mathematical Approach." *Frontiers in Applied Physics and Mathematics* 1, no. 1: 42–56.

Ansari, F. A., H. Hassan, A. S. S. Al-Ouweini, et al. 2026. "An Integrated Algal Biorefinery Approach for Wastewater Treatment and Biomass Valorisation." *Sustainability* 18, no. 4: 2123.

Araújo, S. F. A., A. Á. M. Ferreira, L. A. S. de Oliveira, et al. "Chlorella Vulgaris Biorefineries: Sustainable Biofuels and High-Value Carbon Capture." *Biofuels, Bioproducts and Biorefining*, (2026).

Arulrajah, P., A. E. Lievonen, D. Subaşı, S. Pagal, D. Weuster-Botz, and A. L. Heins. 2025. "Scale-Down Bioreactors—Comparative Analysis of Configurations." *Bioprocess and Biosystems Engineering* 45: 1–17.

Bedru, T. K., W. B. Garuma, R. Baskaran, M. Jayakumar, and T. Selvaganapathy. 2025. "Road Map From Microalgae Biomass to Biorefinery Prospects and Challenges." In *Algal Biorefineries in Circular Economy for Climate Change Resilience*, edited by F. A. Malla, and S. A. Bandh, 57–80. Climate Change Resilience Press.

Budi Susilo, Y. K., K. Loganathan, P. Sukirthanandan, and S. Abdul Rahman. 2025. "Sustainable Biofuel From Algae: An Artificial Intelligence Optimization Approach for Production Enhancement." *Sustainable Biofuel from Algae: An Artificial Intelligence Optimization Approach for Production Enhancement*.

Butean, A., I. Cutean, R. Barbero, J. Enriquez, and A. Matei. 2025. "A Review of Artificial Intelligence Applications for Biorefineries and Bioprocessing: From Data-Driven Processes to Optimization Strategies and Real-Time Control." *Processes* 13, no. 8: 2544.

Coşgun, A., M. E. Günay, and R. Yıldırım. 2023. "Machine Learning for Algal Biofuels: A Critical Review and Perspective for the Future." *Green Chemistry* 25, no. 9: 3354–3373.

Costantini, L., 2025. "Assessment of Tools and Methodologies for LCA and TEA of Advanced Biofuels: A Preliminary Case Study on the SUNPERFORM Project." Doctoral diss., Politecnico di Torino.

Deepankumar, S. 2025. "Enhancement of CI Engine Performance and Emission Characteristics Using SiO_2 Nanoparticle-Enriched Biofuel Derived From Waste Paint and Prosopis Juliflora." *Silicon* 17: 1–20.

Deepankumar, S., and S. K. Kl. 2026. "Hydrothermal Liquefaction and Gasification of Industrial Waste Algae: Experimental and AI-Assisted Optimization for Biofuel and Hydrogen Production." *Biomass Conversion and Biorefinery* 16, no. 3: 125.

Demirel, Y., and M. A. Rosen. 2023. *Sustainable Engineering: Process Intensification, Energy Analysis, and Artificial Intelligence*. CRC Press.

Deprá, M. C., R. R. Dias, L. Q. Zepka, and E. Jacob-Lopes. 2024. "Algae for Environmental Sustainability: Trends and Future Outlook." In *Algae as a Natural Solution for Challenges in Water-Food-Energy Nexus: Toward Carbon Neutrality*. edited by T. A. Kurniawan, and A. Anouzla, 861–869. Springer Nature Singapore.

Dubey, D., and V. Dutta. 2025. "Use of Artificial Intelligence for Monitoring Algal Blooms in Aquatic Ecosystem." In *Artificial Intelligence in Microbial Research: Bridging the Gap*. edited by B. Pandey, D. Pandey, A. Khamparia, V. Dutta, and V. E. Bala, 101–127. Springer Nature Singapore.

Freitas, G. R., S. Badenes, R. Oliveira, and F. G. Martins. 2025. "A Review of Intelligent Modeling for Microalgae Systems: Integrating Data Mining, Machine Learning, and Hybrid Approaches." *Processes* 13, no. 9: 2956.

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Ge, Y., K. Ying, G. Yu, et al. 2025. "A Systematic Review on Machine Learning-Aided Design of Engineered Biochar for Soil and Water Contaminant Removal." *Frontiers in Soil Science* 5: 1623083.

Geng, Y., A. Shaukat, W. Azhar, et al. 2025. "Microalgal Biorefineries: A Systematic Review of Technological Trade-Offs and Innovation Pathways." *Biotechnology for Biofuels and Bioproducts* 18, no. 1: 93.

Gerdan Koc, D., C. Koc, and K. Ekinci. 2023. "Fusion-Based Machine Learning Approach for Classification of Algae Varieties Exposed to Different Light Sources in the Growth Stage." *Algal Research* 71: 103087.

Giacomella, L. 2021. "Techno-Economic Assessment (TEA) and Life Cycle Costing Analysis (LCCA): Discussing Methodological Steps and Integrability." *Insights Into Regional Development* 3, no. 2: 176–197.

Gu, X., L. Yu, N. Pang, J. S. Martinez-Fernandez, X. Fu, and S. Chen. 2020. "Comparative Techno-Economic Analysis of Algal Biofuel Production via Hydrothermal Liquefaction: One Stage Versus Two Stages." *Applied Energy* 259: 114115.

Guo, B., X. Lu, X. Jiang, X. L. Shen, Z. Wei, and Y. Zhang. 2025. "Artificial Intelligence in Advancing Algal Bioactive Ingredients: Production, Characterization, and Application." *Foods* 14, no. 10: 1783.

Gupta, S., V. Gautam, and S. Kanwar. 2025. "Machine Learning in Biotechnology: Current Applications and Future Prospects." *From Genes to Algorithms: Navigating the Biotechnology Data Revolution*, edited by P. Bhambri, S. Kautish, and Y. Gupta, 21–40.

Hamood ur Rehman, M., M. Hussain, P. Akhter, and F. Jamil. 2025. "Breaking Barriers in Biodiesel: From Feedstock Challenges to Technological Advancements." *Chemistry Africa* 8: 2225–2256.

Haq, M. A. U., M. Irfan, S. Malik, et al. 2026. "Machine Learning–Enabled Microalgae Bioprocess Optimization for Alternative Foods and Water Sustainability." *Current Pollution Reports* 12, no. 1: 23.

He, L., A. A. I. Luthfi, N. M. Saleh, et al. 2025. "Microalgae Biodiesel: Advances, Costs, and Potential." *Waste and Biomass Valorization* 17: 1–19.

Hosseini, M., S. J. Amirfakhri, and R. Ghiaasiaan. 2025. "Artificial Intelligence in Biofuels: Progress, Trends, and Directions." *Chemistry & Biodiversity*: e00430.

Imamoglu, E. 2024. "Artificial Intelligence and/or Machine Learning Algorithms in Microalgae Bioprocesses." *Bioengineering* 11, no. 11: 1143.

Inamdar, M. A., A. Gudigar, U. Raghavendra, et al. 2026. "TrustNet: A Lightweight Network With Integrated Uncertainty Quantification and Quantitative Explainable AI for Ischemic Stroke Detection in CT Images." *Scientific Reports* 16: 9861.

Jathar, L., M. Kolhe, J. Patil, and U. Awasarmol. 2025. "Harnessing Microalgae for Sustainable Biofuel: Integration of Biochemical Conversion, AI/ML Applications and Life Cycle Analysis." *International Journal of Ambient Energy* 46, no. 1: 2543914.

Jayabal, R., and P. Prabhakar. 2025. "Advanced Strategies in Food and Agricultural Waste Valorisation: Integrating Nanoengineering, Artificial Intelligence and Smart Bioconversion for Sustainable Resource Recovery." *Journal of the Science of Food and Agriculture* 106, no. 8: 4507–4520.

Jena, S. R., and S. Agarwal. 2026. "AI-Based Sustainable Green Energy Optimization for Edge Cloud Computing With Renewable Energy Resources." In *Artificial Intelligence and Sustainable Innovation*. edited by A. Dagur, S. Agarwal, D. Kumar Shukla, S. Ali, and S. Sharma, 212–219. CRC Press.

Karaca, Y., and D. Baleanu. 2022. "Evolutionary Mathematical Science, Fractional Modeling and Artificial Intelligence of Nonlinear Dynamics in Complex Systems." *Chaos Theory and Applications* 4, no. 3: 111–118.

Karthikeyan, B., P. Kumar, P. Karthikeyan, et al. 2026. "Adaptive artificial intelligence for intelligent fault detection and correction in sensor systems." In *Adaptive AI in Sensor Informatics*. edited by K. Ramamurthy, S. Kulanthaivelu, and S. Kadry, 133–152. Elsevier.

Khan, A., S. Ullah, R. Ali, et al. 2025. "Machine Learning Optimization of Environmental Factors Influencing Biomass and Nutritional Composition in Local Algal Species." *Royal Society Open Science* 12, no. 4: 241336.

Krejcar, O., J. Abdullah, and H. Namazi. 2026. "Implementing XAI in Life Sciences: Key Challenges and Pathways to Solutions." *Artificial Intelligence in the Life Sciences* 9: 100153.

Kuila, A. and and Kumar, D., ed. 2025. *Optimizing Biofuel Production with Artificial Intelligence*. John Wiley & Sons.

Kumar, R. R., D. Sarkar, and R. Sen. 2024. "Simultaneously Maximizing Microalgal Biomass and Lipid Productivities by Machine Learning Driven Modeling, Global Sensitivity Analysis and Multi-Objective Optimization for Sustainable Biodiesel Production." *Applied Energy* 358: 122597.

Kushwaha, O. S., H. Uthayakumar, and K. Kumaresan. 2023. "Modeling of Carbon Dioxide Fixation by Microalgae Using Hybrid Artificial Intelligence (AI) and Fuzzy Logic (FL) Methods and Optimization by Genetic Algorithm (GA)." *Environmental Science and Pollution Research* 30, no. 10: 24927–24948.

Lin, Y., J. Tang, J. Guo, S. Wu, and Z. Li. 2025. "Advancing AI-Enabled Techniques in Energy System Modeling: A Review of Data-Driven, Mechanism-Driven, and Hybrid Modeling Approaches." *Energies* 18, no. 4: 845.

Madika, B., A. Saha, C. Kang, et al. 2025. "Artificial Intelligence for Materials Discovery, Development, and Optimization." *ACS Nano* 19, no. 30: 27116–27158.

Mafat, I. H., S. Palla, and D. V. Surya. 2024. "Machine Learning and Artificial Intelligence for Algal Cultivation, Harvesting Techniques, Wastewater Treatment, Nutrient Recovery, and Biofuel Production and Optimization." In *Value Added Products From Bioalgae Based Biorefineries: Opportunities and Challenges*. edited by S. Kumar Arya, M. Khatri, and G. Singh, 463–487. Springer Nature Singapore.

Malashin, I. P., D. Martysyuk, V. Nelyub, A. Borodulin, A. Gantimurov, and V. Tynchenko. 2026. "Multi-Objective Optimisation in Biofuel and Bioenergy Systems: A Comprehensive Review." *International Journal of Sustainable Engineering* 19, no. 1: 2619227.

Malik, H. A., R. Shahid, O. Rehman, et al. 2026. "Development of Algal Industry 4.0 for Efficient Cultivation, Resource Recovery, and Bioproducts Production." In *Algae Biotechnology*. edited by M. A. Mehmood, S. Malik, and M. J. Betenbaugh, 355–374. Elsevier.

Maneedaeng, A., A. Wiangkham, A. Ariyarit, A. Pumpuang, and E. Sukjit. 2025. "Investigating the Impact of Fatty Acid Profiles on Biodiesel Lubricity Using Artificial Intelligence Techniques." *Cleaner Engineering and Technology* 25: 100913.

Maroušek, J., A. Marouškovᡠ, B. Gavurová, D. Tuček, and O. Strunecký. 2023. "Competitive Algae Biodiesel Depends on Advances in Mass Algae Cultivation." *Bioresource Technology* 374: 128802.

Meenatchisundaram, K., S. C. Gowd, J. Lee, S. Barathi, and K. Rajendran. 2024. "Data-Driven Model Development for Prediction and Optimization of Biomass Yield of Microalgae-Based Wastewater Treatment." *Sustainable Energy Technologies and Assessments* 63: 103670.

Metodiev, M., A. Dexter, W. Zhou, et al. 2026. "Evaluating Batch Correction Methods for Large-Scale Mass Spectrometry Imaging of Heterogeneous Tissues." *Analytical Chemistry* 98: 3531–3543.

Mir, B. A., R. Kumari, A. D. Raju, et al. 2025. "Artificial Intelligence-Based Microalgal Biofuel Production: Future Prospects, Limitations, and Challenges." In *Artificial Intelligence for Biomass-based Biofuel Production*. edited by A. Kuila, and A. Shastri, 254–277. CRC Press.

Mirakhori, F., and S. K. Niazi. 2025. "Harnessing the AI/ML in Drug and Biological Products Discovery and Development: The Regulatory Perspective." *Pharmaceuticals* 18, no. 1: 47.

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Mohapatra, A., E. Guliani, V. Nandal, et al. 2025. "Recent Innovations in Sustainable Algal Biofuel Production: A Comprehensive Review." *Brazilian Journal of Chemical Engineering* 42: 1–29.

Molla, A., and G. Meseret. 2025. "Strain Improvement Through Genetic Engineering and Synthetic Biology for the Creation of Microalgae With Enhanced Lipid Accumulation, Stress Tolerance, and Production of High-Value." *Science* 6, no. 3: 80–95.

Obidi, P. O., and D. J. Bayless. 2025. "The Synergy of Artificial Intelligence and Algal Systems for Sustainable Wastewater Treatment and Carbon Sequestration." *Environmental Technology Reviews* 14, no. 1: 1040–1067.

Omole, O. A., C. C. Ogbaga,J. A., Okolie, O. Akande, R. Kimera, R. and J. L., Dayil. 2025. "Advancing Algal Biofuel Production Through Data-Driven Insights: A Comprehensive Review of Machine Learning applications." *Computers & Chemical Engineering* 196: p.109049.

Onay, M., and A. Onay. 2026. "Integrated Microalgal Biorefineries for Energy and Food Production." In *Handbook of Microalgae-Based Processes and Products*. edited by E. Jacob-Lopes, M. M. Maroneze, M. I. Queiroz, and L. Q. Zepka, 899–913. Academic Press.

Onianwah, I. F., I. V. Chukuka, M. O. Adeola, and O. F. Onoriasakpobare. 2025. "The Application of Artificial Intelligence and Machine Learning in Optimizing Processes in Algal Technology: Future Characteristics and Opportunities." *Journal of Advanced Science and Optimization Research* 8, no. 9: 141–156.

Oruganti, R. K., A. P. Biji, T. Lanuyanger, et al. 2023. "Artificial Intelligence and Machine Learning Tools for High-Performance Microalgal Wastewater Treatment and Algal Biorefinery: A Critical Review." *Science of the Total Environment* 876: 162797.

Park, J., K. Patel, and W. H. Lee. 2024. "Recent Advances in Algal Bloom Detection and Prediction Technology Using Machine Learning." *Science of The Total Environment* 938: 173546.

Pasha, M. K., L. Dai, D. Liu, M. Guo, and W. Du. 2021. "An Overview to Process Design, Simulation and Sustainability Evaluation of Biodiesel Production." *Biotechnology for Biofuels* 14, no. 1: 129.

Patel, A. K. 2025. "Microalgae in Dual Role: An Effective Platform for Biological Carbon Capturing and High-Value Carotenoid Production." *Current Pollution Reports* 11, no. 1: 34.

Pavendan, K., and V. Nagarajan. 2022. "Modelling of Wastewater Treatment, Microalgae Growth and Harvesting by Flocculation Inside Photo Bioreactor Using Machine Learning Technique." *Journal of Intelligent & Fuzzy Systems* 43, no. 5: 5607–5620.

Privat, E., I. Fortunati, C. Ferrante, S. Rampino, and A. Polimeno. 2025. "Data-Driven Analysis of Fluorescence Lifetime Imaging Experiments: Unraveling the Signal/Stress Relationship of Polluted Microalgae Cells With Machine Learning." *ACS Omega* 10, no. 23: 25111–25119.

Pusuluru, S., K. Velmurugan, S. K. Punna, et al. 2026. "Advancing Biofuel Production Through Algal Biorefineries: Optimization of Polysaccharide Utilization, AI-Driven Innovations, and Sustainability Assessments." *Biomass and Bioenergy* 204: 108366.

Qian, J., L. Qian, N. Pu, Y. Bi, A. Wilhelms, and S. Norra. 2024. "An Intelligent Early Warning System for Harmful Algal Blooms: Harnessing the Power of Big Data and Deep Learning." *Environmental Science & Technology* 58, no. 35: 15607–15618.

Rangel Pinto, J. D., J. L. Guerrero, L. Rivera, et al. 2024. "Predicting the Microalgae Lipid Profile Obtained by Supercritical Fluid Extraction Using a Machine Learning Model." *Frontiers in Chemistry* 12: 1480887.

Rayamajhi, V., M. Hussain, H. Shin, and S. Jung. 2025. "Recent Advances in the Application of Artificial Intelligence in Microalgal Cultivation." *Processes* 13, no. 12: 3764.

Riezzo, L., H. Kay, Y. Feng, K. Jing, and D. Zhang. 2025. "Accelerating Bioprocess Digital Twin Development by Integrating Hybrid Modelling With Transfer Learning." *Chemical Engineering Journal* 511: 162018.

Saju, L., D. Selvaraj, and T. Vairaperumal. 2025. "Artificial Intelligence and Machine Intelligence: Modeling and Optimization of Bioenergy Production." In *Computer Vision and Machine Intelligence for Renewable Energy Systems*. edited by A. Kumar Dubey, A. Lal Srivastav, and V. García-Díaz, 163–176. Elsevier.

Sampaio, I. C. F., A. S. da Silva Quinto, I. V. L. de Moura, K. S. Cezar, M. Franco, and P. F. de Almeida. 2024. "The Role of Composting in Sustainable Agriculture and Waste Management." In *Progress in Hydrogen Energy, Fuel Cells, Nano-Biotechnology and Advanced, Bioactive Compounds*. edited by C. A. Taft, and S. R. de Lazaro, 417–432. Springer Nature Switzerland.

Santos, B., F. Freitas, A. J. F. N. Sobral, and T. Encarnação. 2025. "Microalgae and Circular Economy: Unlocking Waste to Resource Pathways for Sustainable Development." *International Journal of Sustainable Engineering* 18, no. 1: 2501488.

Saravanakumar, K., K. Sathyanarayanan, M. S. Khan, et al. 2024. "Recent Progress in Biotechnological Approaches for Diverse Applications of Algae: An Overview." *International Journal of Environmental Science and Technology* 21, no. 3: 3453–3474.

Sarkar, S., K. Gayen, and T. K. Bhowmick. 2025. "Green Extraction of Biomolecules From Algae Using Subcritical and Supercritical Fluids." *Biomass Conversion and Biorefinery* 15, no. 15: 21603–21625.

Satish Kumar, R., B. N. Rao, M. Prameela, et al. 2022. "Assessment of Bioprocess Development-Based Modeling and Simulation in a Sustainable Environment." *International Journal of Photoenergy* 2022, no. 1: 1–10.

Semwal, S., and H. C. Joshi. 2026. "Technobiological Pathways for High-CO_2 Capture Using Micro-/Macroalgae: Genetic Engineering, Process Automation, and Value-Added Bioproducts." *Asia-Pacific Journal of Chemical Engineering* 21: e70225.

Sharma, M., K. Goria, V. V. Tyagi, K. Chopra, and R. Kothari. 2026. "Sustainability Assessment of Open Bioreactors for Algal Biomass With Modelling." *Algal Photobioreactors*: 133–164.

Sharma, S., K. Sharma, and S. Grover. 2024. "Real-Time Data Analysis With Smart Sensors." In *Application of Artificial Intelligence in Wastewater Treatment*, edited by S. Gulati, 127–153. Springer Nature Switzerland.

Sheik, A. G., A. Kumar, F. A. Ansari, et al. 2024. "Reinvigorating Algal Cultivation for Biomass Production With Digital Twin Technology-A Smart Sustainable Infrastructure." *Algal Research* 84: 103779.

Sheik, A. G., A. Kumar, R. Patnaik, S. Kumari, and F. Bux. 2024. "Machine Learning-Based Design and Monitoring of Algae Blooms: Recent Trends and Future Perspectives–A Short Review." *Critical Reviews in Environmental Science and Technology* 54, no. 7: 509–532.

Shokravi, Z., and N. Mehravar. 2025. "Optimization of Photobioreactor and Open Pond Systems for Sustainable Microalgal Biomass Production: Challenges, Solutions, and Scalability." In *Microalgal Biofuels*, edited by K. Kumar Jaiswal, B. Singhand A. K. Jaiswal, 45–62. Woodhead Publishing.

Sierra-Espinosa, F. Z., F. Carrillo, and D. M. Graciano. 2025. "Computational Analysis and Modeling of Algal Bioreactors Performance." In *Algal Bioreactors.*, Edited by Y. Gagnon, E. Jacob-Lopes, M. Costa Deprá et al., 189–204. Elsevier Science Ltd.

Singh, A., and P. Sharma. 2025. "Deep Learning Applications in Algal Biotechnology: Current Status and Future Perspectives." *Algal Research* 78: 103345.

Sirohi, R., Y. J. Sung, S. B. Ummalyma, A. Vadiveloo, H. C. Yadav, and A. Tarafdar. 2026. "Machine Learning Approaches for Smart Cultivation and Processing of Microalgae." *Systems Microbiology and Biomanufacturing* 6, no. 2: 42.

Slade, R., and A. Bauen. 2013. "Micro-Algae Cultivation for Biofuels: Cost, Energy Balance, Environmental Impacts and Future Prospects." *Biomass and Bioenergy* 53: 29–38.

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Subburaj, S., C. Liu, and T. Xu. 2025. "Emerging Trends in AI-Integrated Optical Biosensors for Point-of-Care Diagnostics: Current Status and Future Prospects." *Chemical Communications* 61: 18464–18489.

Sultana, N., S. M. Z. Hossain, M. Abusaad, N. Alanbar, Y. Senan, and S. A. Razzak. 2022. "Prediction of Biodiesel Production From Microalgal Oil Using Bayesian Optimization Algorithm-Based Machine Learning Approaches." *Fuel* 309: 122184.

Syed, T., F. Krujatz, Y. Ihadjadene, et al. 2024. "A Review on Machine Learning Approaches for Microalgae Cultivation Systems." *Computers in Biology and Medicine* 172: 108248.

Tang, D. Y. Y., C. Ooi, and P. L. Show. 2025. "Blue Bioeconomy and Biotechnology: Towards a Sustainably Growing Microalgae Industry." *Blue Biotechnology* 2, no. 1: 13.

Tang, J., Z. Zhang, J. Wang, F. Kong, and Z. Chi. 2026. "Machine Learning-Driven Multi-Objective Optimization of Dunaliella Salina Cultivation for Enhanced Biomass and β-Carotene Production." *Bioresource Technology* 446: 134182.

Tripathi, G., P. Dubey, A. Shamim, A. Farooqui, and V. Mishra. 2024. "Bio-Flocculation: A Cost Effective and Energy Efficient Harvesting Technique for Algal Biofuel Production and Wastewater Treatment." *Bioresource Technology Reports* 28: 101969.

Tripathi, G., P. Dubey, K. Osama, et al. 2025. "Two-Stage Cultivation of Microalga for Sustainable Biofuel Production." *Green Chemistry Letters and Reviews* 18, no. 1: 2543909.

Tsopanoglou, A., and I. Jiménez del Val. 2021. "Moving Towards an Era of Hybrid Modelling: Advantages and Challenges of Coupling Mechanistic and Data-Driven Models for Upstream Pharmaceutical Bioprocesses." *Current Opinion in Chemical Engineering* 32: 100691.

Tummawai, T., T. Rohitatisha Srinophakun, S. Padungthon, and S. Sukpancharoen. 2024. "Application of Artificial Intelligence and Image Processing for the Cultivation of Chlorella sp. Using Tubular Photobioreactors." *ACS Omega* 9, no. 46: 46017–46029.

Udaypal, U., R. K. Goswami, S. Mehariya, O. P. Karthikeyan, and P. Verma. 2026. "Open Photobioreactor Design and Operational Strategies for Algae Cultivation: Challenges and Solutions." In *Algal Photobioreactors*, 289–308. Springer Nature Singapore.

Vaughan, L., A. Zamyadi, S. Ajjampur, H. Almutaram, and S. Freguia. 2022. "A Review of Microscopic Cell Imaging and Neural Network Recognition for Synergistic Cyanobacteria Identification and Enumeration." *Analytical Sciences* 38, no. 2: 261–279.

Venkatrayalu, G., S. Choudhary, and K. M. Poluri. 2026. "AI-Assisted Microalgal Bioprocessing: Innovations, Challenges, and Future Prospects in Sustainable Manufacturing." In *Industrial Microbiology and Biotechnology: Trends in Microbial Biotechnology*, edited by R. Kothari, P. Verma, and V. Veer Tyagi, 269–307.

Wu, W., C. M. Huang, and Y. H. Tsai. 2023. "Design and Validation of a Microalgae Biorefinery Using Machine Learning-Assisted Modeling of Hydrothermal Liquefaction." *Algal Research* 74: 103230.

Wu, Y., L. Shan, W. Zhao, and X. Lu. 2025. "Harnessing Artificial Intelligence to Revolutionize Microalgae Biotechnology: Unlocking Sustainable Solutions for High-Value Compounds and Carbon Neutrality." *Marine Drugs* 23, no. 5: 184.

Xian, Y., C. Li, Y. Xu, Y. Zhou, and D. Xue. 2025. "AI-Driven Advances in Sustainable Materials for Green Energy: From Innovation to Lifecycle Management." *SusMat* 5, no. 5: e70030.

Xie, J., and Q. Zhao. 2025. "Green Extraction of Active Compounds in Marine Microalgae." *Blue Biotechnology* 2, no. 1: 30.

Xu, K., Z. Zhu, H. Yu, and X. Zou. 2025. "Prediction of Microalgae Harvesting Efficiency and Identification of Important Parameters for Ballasted Flotation Using an Optimized Machine Learning Model." *Algal Research* 87: 103985.

Xue, Z., S. Li, W. Yu, et al. 2021. "Research Advancement and Commercialization of Microalgae Edible Oil: A Review." *Journal of the Science of Food and Agriculture* 101, no. 14: 5763–5774.

Yaashikaa, P. R., M. Keerthana Devi, P. Senthil Kumar, and E. Pandian. 2022. "A Review on Biodiesel Production by Algal Biomass: Outlook on Lifecycle Assessment and Techno-Economic Analysis." *Fuel* 324: 124774.

Yadav, J., H. Marwah, and C. Kumar. 2025. "Synthetic Biology and Metabolic Engineering Paving the Way for Sustainable Next-Gen Biofuels: A Comprehensive Review." *Energy Advances* 4, no. 10: 1209–1228.

Yahaya, E., W. S. Yeo, J. Nandong, and J. C. Y. Ngu. 2025. "CO_2 Fed Microalgae Cultivation in Photobioreactor: Review on Challenges and Possible Solutions." *Environmental Technology Reviews* 14, no. 1: 540–564.

Yang, C. T., E. Kristiani, Y. K. Leong, and J. S. Chang. 2024. "Machine Learning in Microalgae Biotechnology for Sustainable Biofuel Production: Advancements, Applications, and Prospects." *Bioresource Technology* 413: 131549.

Yang, J., M. Xu, X. Zhang, Q. Hu, M. Sommerfeld, and Y. Chen. 2011. "Life-Cycle Analysis on Biodiesel Production From Microalgae: Water Footprint and Nutrients Balance." *Bioresource technology* 102, no. 1: 159–165.

Yang, Y., X. Chen, J. Wang, Y. Dong, K. Qie, and Z. Yuan. 2026. "Enhancing Vision-Based Traffic Crash Detection Performance Consistency Across Day-Night Scenes: A Depth-Aware and Domain-Adaptive Network." *Accident; Analysis and Prevention* 228: 108405.

Yeh, Y. C., T. Syed, G. Brinitzer, et al. 2023. "Improving Microalgae Growth Modeling of Outdoor Cultivation With Light History Data Using Machine Learning Models: A Comparative Study." *Bioresource Technology* 390: 129882.

Yusof, Z., Y. W. Tong, K. Selvarajoo, S. K. Parakh, and S. C. Foo. 2025. "Overcoming Challenges in Microalgal Bioprocessing Through Data-Driven and Computational Approaches." *Current Opinion in Food Science* 61: 101253.

Yusuff, A. S., N. B. Ishola, A. O. Gbadamosi, T. M. Azeez, and M. O. Onibonoje. 2023. "An Artificial Intelligence Approach to Model and Optimize Biodiesel Production From Used Cooking Oil Using CaO Incorporated Zeolite Catalyst." *Energy Conversion and Management: X* 20: 100452.

Zahra, E., M. Manish, J. Baltrusaitis, D. Pallavi, and M. W. Mark. 2025. "Emerging Trends in Sustainable Energy System Assessments: Integration of Machine Learning With Techno-Economic Analysis and Lifecycle Assessment." *Sustainability Science and Technology* 2, no. 1: 012001.

Zaman, H. B., K. L. Yu, H. Mohamed, P. J. Ker, and M. N. I. Salehmin. 2025. "Integration of Energy Informatics in Microalgae Biorefinery and Bioenergy Towards Environmental and Energy Sustainability in Smart Cities." In *International Visual Informatics Conference*. edited by H. B. Zaman, P. Robinson, A. F. Smeaton, et al., 434–446. Springer Nature Singapore.

10970290, 0, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/bit.70305 by Faculdade Medicina De Lisboa, Wiley Online Library on [03/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License